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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION

October 2026

Fifth SEMESTER

DSE-MG5DSEBOT304 - ANALYTICAL TECHNIQUES IN PLANT SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Part A

Very Short Answer Questions. Answer any **10** questions. Each question carries **2** marks

1. What is glycerine and what is its use in microtechnique? [U/CO1]

Glycerine is a colourless, viscous, hygroscopic trihydric alcohol **(1 mark).** Use: serves as a temporary mounting medium, prevents drying of sections, softens and clears tissues. **(1 mark).**

2. Difference between sledge and rotary microtome. [K/CO1]

Sledge microtome: The knife moves over a fixed specimen. Used for hard or woody material and gives thick sections. Rotary microtome: the knife is fixed and the specimen block moves up and down by a rotating hand-wheel; used for paraffin-embedded material and gives thin serial sections/ribbons.

3. Define the smear technique in plant cytological studies. [K/CO1]

A technique in which cells are spread as a thin film on a slide by gentle pressing or spreading.

4. Define maceration in plant microtechnique. [K/CO1]

Maceration is the process of separating the cells of a tissue by dissolving the middle lamella with chemicals (e.g. Schultze's method, Jeffrey's method) so that individual cells such as vessels, tracheids and fibres can be studied.

5. State the two basic methods of preserving plant specimens. [K/CO1]

(i) Dry preservation

(ii) Wet preservation

Or Museum specimen/Herbarium

6. Name different types of fixation. [K/CO1]

Physical fixation and chemical fixation

7. What is meant by Chamber volume in haemocytometer counting. [K/CO4]

Liberal valuation

The volume of liquid held between the chamber floor and the coverslip over a defined counting area. Volume calculation- For a standard 1 mm² large square: 1 mm² × 0.1 mm = 0.1 mm³ (0.1 μL).

8. Mention any two applications of UV-Visible spectroscopy. [U/CO5]

Quantitative estimation; Determination of concentration, enzyme activity assays; purity checking **(Any two**).

9. How does particle size affect sedimentation? [A/CO4]

Larger particles sediment faster; smaller particles sediment slowly and need higher centrifugal force.

10. Describe the Principle of lyophilizer. [U/CO4]

Freeze-drying: the material is frozen and placed under high vacuum, so the ice sublimes directly to vapour without melting, and the vapour is trapped on a cold condenser.

11. Which electrode do DNA molecules migrate toward during agarose gel electrophoresis?[U/CO2]

The anode (positive electrode)

12. What are the different components of TLC. [U/CO5]

Stationary phase (silica gel/alumina layer on glass or aluminium plate with binder); mobile phase (solvent system); developing chamber with lid; sample applicator (capillary); detection/visualisation (UV, iodine vapour, spray reagents); **(Any Two)**

Part B

Short Answer Questions.Answer any **8** questions. Each question carries **5** marks

13. Explain the preparation and use of safranin in plant microtechnique. [U/CO1]

Liberal valuation

In syllabus, it's mentioned to study only the composition. Preparation:

- Stock solution- Add 2.5 g safranin-O to 100.0 ml 95% ethyl alcohol.
- Working solution-Add 10.0 ml safranin and ethyl alcohol solution made in step 1 to 90.0 ml distilled water. (2.5 marks).

Use: stains lignified, suberised and cutinised walls (xylem, sclerenchyma), nuclei and chromosomes red; commonly used with fast green/light green as counterstain in double staining of sections (2.5 marks).

14. Describe the general classification of fixatives with suitable examples. [U/CO1]

- Based on composition: simple fixatives (single chemical: ethanol, formaldehyde, acetic acid, mercuric chloride, osmium tetroxide) and compound fixatives (mixtures: FAA, Carnoy's, Bouin's, Navashin's, Farmer's fluid).
- Based on action: coagulant (ethanol, mercuric chloride, picric acid) and non-coagulant (formaldehyde, acetic acid, osmium tetroxide).
- Based on purpose: micro-anatomical, cytological fixatives and Histochemical
- Physical and chemical

15. Working of a compound light microscope. [U/CO3]

- Parts: mechanical (base, arm, stage, body tube, nosepiece, focusing knobs) and optical (light source/mirror, condenser and diaphragm, objectives, eyepiece).
- Working: light is focused by the condenser on the specimen; the objective forms a real, inverted, magnified primary image; the eyepiece further magnifies it to a virtual image. Total magnification = objective × eyepiece power.

16. Explain the Principle and working of a spectrophotometer. [U/CO2]

- Principle (2 marks): Beer–Lambert law; absorbance is proportional to concentration and path length.
- Working (2.5 marks): monochromatic light passes through the blank and then the sample; the detector measures transmitted light, converted to absorbance; concentration is read from a standard curve.

17. Explain structural components of a pH meter. [U/CO3]

Any five points, 1 mark each: glass electrode (thin glass bulb sensitive to H^+, Ag/AgCl wire in 0.1 M HCl); reference electrode (calomel or Ag/AgCl with KCl); combination electrode; high-impedance millivoltmeter/amplifier; temperature probe/ATC; digital display; calibration knobs (standard buffers pH 4, 7, 10).

18. Describe the process of Image formation in SEM. [U/CO3]

- Electron gun produces a beam (tungsten filament/FEG, 1–30 kV), focused by condenser and objective lenses under high vacuum.
- Scanning the specimen-Special scanning coils sweep the probe back and forth across the coated specimen surface.
- Beam–specimen interaction yields secondary electrons (topography), backscattered electrons (composition) and X-rays.
- The detector collects the signal, which is amplified and synchronised with the scan to build the image on a monitor.
- Sample preparation (optional): fixation, dehydration, critical point drying, gold sputter coating.

19. Explain the Principle and applications of ultracentrifugation. [U/CO2]

- Principle: separation of particles by sedimentation under very high centrifugal force (up to about $10^6 \times g$, above 60,000 rpm) in a refrigerated vacuum chamber; rate depends on size, shape, density (sedimentation coefficient, Svedberg unit S); $RCF = 1.118 \times 10^{-5} \times r \times rpm^2$.
- Applications: isolation of organelles, ribosomes, viruses; purification of DNA/plasmids and proteins; determination of molecular weight and density, etc.

20. How the electric charge of a molecule affects electrophoretic mobility. [U/CO2]

- A molecule moves in an electric field because it has an electrical charge.
- The greater its net charge, the more strongly it is attracted towards the oppositely charged electrode and the faster it moves.
- positively charged molecules move to the cathode and negatively charged ones to the anode
- Net charge of a molecule depends on its chemical groups, and the pH of the surrounding buffer

21. Briefly explain the working and application of paper chromatography. [U/CO3]

Working:

- Put a small spot of the sample on the bottom of the paper, and let it dry.
- Hang the Whatman filter paper (Stationary phase) in a closed chamber with the lower edge dipping in the solvent (mobile phase), (below the spot).
- The solvent rises by capillary action and carries the compounds with it. Compounds that dissolve better in the solvent move faster and further.
- Remove the paper when the solvent front nears the top and mark the front. Dry the paper.
- Calculate Rf = distance travelled by the compound ÷ distance travelled by the solvent front.

Applications - Separation and identification of amino acids and sugars, Separation of plant pigments such as chlorophylls and carotenoids, Analysis of flavonoids, organic acids and other plant metabolites, Checking the purity of a compound

22. Compare and contrast HPLC and column chromatography. [U/CO5]

Award 5 marks for a comparison covering at least five points, plus similarities.

Feature	Column chromatography	HPLC
Stationary phase	Packed column	Fine packed column
Driving Force	Gravity flow	High pressure pump
Particle size	Large (60–200 μm)	Small (3–10 μm)
Resolution	Low	High

Detection	Manual fractions, offline	Online detectors (UV, PDA, RI, fluorescence)
Sample amount	Large	Small
Scale	preparative	analytical or preparative
Speed	Slow	Fast
Cost	Low	High (pump, column, detector)

Similarity: both use a stationary and a mobile phase and separate by differential adsorption/partition (1 mark).

Part C

Essay Questions. Answer any 1 questions. Each question carries 10 marks

23. Distinguish between temporary and permanent slide preparation techniques. Discuss the steps involved and suitable mounting media for each. [U/CO1]

- Distinction (3 marks): temporary slides are quick, last only a short time, mounted in water/glycerine; permanent slides are made from fixed, processed material, stored for years, mounted in a resinous medium. (Include at least three points of difference.)
- Steps in temporary preparation (2 marks): select material, cut thin freehand section, stain (safranin/toluidine blue, iodine), mount in water or dilute glycerine.
- Steps in permanent preparation (3 marks): killing and fixing (FAA), dehydration (graded alcohol series), clearing (xylene), embedding in paraffin wax, sectioning (rotary microtome), staining (e.g. safranin–fast green), mounting
- Mounting media (2 marks): temporary: water, dilute glycerine, lactophenol; semi-permanent: glycerine jelly; permanent: Canada balsam (RI about 1.52), DPX, Euparal.

24. Explain the working and applications of colorimeter. [U/CO3]

- Components (3 marks): light source (tungsten lamp), filter (selects wavelength range), cuvette holding the sample, photocell/photodetector, galvanometer or digital display.
- Working (3 marks): zero with blank, select filter (complementary colour), read absorbance of standards, plot a standard curve, read concentration of the unknown.
- Applications (4 marks): Quantitative estimation of coloured substances in solution. Measurement of microbial growth, Testing water quality/ soil nutrients etc., Estimation of biomolecules such as proteins and carbohydrates, Estimation of pigments and other biological compounds.(Any 4)